# The Remanent Magnetization of the Basic Tertiary Igneous Rocks of Skye, Inverness-shire

**Mohammed Aftab Khan**

(Received 1959 August 17)

*Summary*

The remanent magnetization of the basic igneous rocks of Skye, the hypersthene gabbro of Ardnamurchan, and the layered gabbro of Rhum has been examined. All the rocks possessing stable magnetization were found to be reversely magnetized. This is believed to be due to the Earth's magnetic field being reversed during the half million years or so during which the rocks were formed.

Reasons are given for supposing that the dips of the Skye rocks are formational. The ancient pole position during part of the Lower Miocene is estimated to be 70°·8 (±3°·8) N. Lat., and 159°·6 (±5°·1) E. Long., a position which is significantly different from that of the present geographic pole. It is concluded that there has been relative movement between North-west Europe and the pole since the Lower Miocene.

The directions of magnetization are used in an attempt to correlate some of the lavas and dykes of Skye. It is also tentatively suggested that the rocks of Skye, Rhum and Ardnamurchan are of approximately the same age.

## 1. Introduction

The polar paths deduced from the magnetization directions of rocks, ranging in age from pre-Cambrian to Recent, from different continents have a generally similar form (Irving 1957). The similarity in form of the paths from each continent suggests that polar wandering has occurred, and the fact that the polar path for each continent is different suggests that there has also been some relative movement between the continents. In particular, the separation of the paths for North America and North-west Europe decreased from the Mesozoic onwards and eventually converged some time in the Tertiary. This convergence may be due to the ceasing of relative movement between the two continents. It may also be due to convergence of the pole positions towards the present geographic pole which results in separations in the Upper Tertiary smaller than the errors of individual determinations. To decide if and when relative movement between the two continents ceased it is necessary to have a series of precise pole positions from rocks from as many horizons as possible between the lowest and uppermost Tertiary from both sides of the Atlantic. There are, however, several difficulties in doing this at the present time.

## Table 1

*Tertiary pole positions estimated from the magnetization directions of North-west European rocks*

Northern hemisphere polar coordinates are given; *dp* and *dm* are the semi-axes of the elliptical error area round the pole at a probability of 95 per cent, *dp* in the colatitude direction and *dm* perpendicular to it.

| Age | Formation | Pole positions Lat. | Long. | *dp* | *dm* | References |
|---|---|---|---|---|---|---|
| Plio-pleistocene | Lavas of Central France | 77·9 N | 153·3 E | 15·7 | 20·3 | Roche 1951 |
| Plio-pleistocene | Lavas of South and South-west Iceland | 87·6 N | 146·8 E | 11·5 | 12·6 | Hospers 1953. |
| Mio-pliocene | Lavas of Central France | 73·1 N | 167·3 W | 13·6 | 19·2 | Roche 1951. |
| "Miocene" | Lavas of North and North-west Iceland | 88·6 N | 5·2 E | 9·6 | 10·3 | Hospers 1953. |
| Oligocene | Intrusive of Central France | 73·0 N | 119·0 E | 12·0 | 16·0 | Roche 1950(a) and (b) |
| "Oligocene" | Dykes of North-east England | 86·0 N | 76·0 W | 21·0 | 25·0 | Bruckshaw & Robertson 1949. |
| "Eocene" | Mull lavas | 76·0 N | 162·0 W | 2·5 | 7·6 | Dutt 1955. |
| "Eocene" | Mull intrusives | 72·0 N | 139·0 E | 11·0 | 14·0 | Vincenz 1954. |
| "Eocene" | Arran dykes | 79·0 N | 154·0 E | 12·0 | 15·0 | Leng 1955. |
| "Eocene" | Antrim lavas | 74·0 N | 133·0 E | 6·0 | 8·0 | Hospers & Charlesworth 1954. |
| "Eocene" | Lundy dykes | 73·6 N | 123·5 E | 7·8 | 10·5 | Blundell 1957. |

In Table 1 are given the pole positions and half angles of the oval curves of confidence at $P = 5$ per cent (Fisher 1953, Irving 1956) obtained by earlier workers on rocks of Tertiary age from North-west Europe. It is not possible to arrange the results into any sort of order for the following reasons.

(a) All the rocks studied are igneous, and their dating has so far been done on the scanty fossil content of associated sediments. In the British Isles, Greenland and Iceland, the flora have been used and their interpretation has varied. Thus, the plants from Ardtun in Mull were first studied by Forbes (1851) and dated as Miocene. W. H. Baily (1869) came to the conclusion that the beds were of Lower Miocene age and thought they were the same age as the plants from the inter-basaltic horizon of Ulster in North-east Ireland and the Atanekerdluk formation of East Greenland. Heer (1868) thought that the plants from among the lavas of North and North-west Iceland, Greenland and the British Isles were of Miocene age. Later, however, J. Starkie Gardner (1885a, b) made a more complete study of the plants, and the following is the possible sequence deduced by him:

North and North-west Iceland—Much later than Mull, Ulster and Greenland. Could be as late as Miocene.

Lough Neagh and Bovey Tracey—Covered a long time. Probably Lower Eocene to Middle Miocene.

Ardtun—Younger than Ulster and Greenland, but still Lower Eocene.

Greenland—Younger than Ulster.

Ulster—Equivalent to the Heersian of Gelinden, i.e. Lower Eocene.

13652465b, 1960, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1960.tb00064.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Seward and Holtum (1930) agree with the Eocene age. Hawkes (1938) expressed the view that the Iceland lavas were Eocene or possibly Oligocene in age and argued that this did not necessarily conflict with the fossil evidence.

More recently J. B. Simpson (1951) concluded from pollen studies that the bulk of the Mull lavas are most probably of late Miocene age; in a private communication he has subsequently stated that he thinks it probable that the Skye lavas sampled during the magnetic investigations are also of Miocene age. Starkie Gardner's sequence is not necessarily invalidated by this, but it seems that the beds are all younger than he supposed. However, the situation is not satisfactory.

The Lundy dykes have been stated to be pre-Bovey Tracey beds, i.e. pre-Upper Oligocene in age (Reid 1913), and Blundell (1957) concluded from the similarity of the mean magnetization direction of the dykes with that of the Antrim lavas that they were Eocene. In view of the doubt of the age of the Bovey Tracey beds and the Antrim lavas, this conclusion may have to be revised.

The intrusives of Mull and the dykes of Arran and North-east England are all evidently later than the Mull lavas, but their relative ages are still open to doubt.

The lavas and intrusives of Central France are well dated by means of plants, mammals and insects (Gignoux 1955).

(b) The estimated pole positions, except possibly those from the lavas of Mull and Antrim, are not accurately defined and are therefore of limited use for studies of polar wandering and continental drift.

(c) It is often very difficult to determine the amount by which igneous rocks have been tilted subsequent to their formation. Lavas are not always laid down horizontally, and when no sediments, half-filled vesicles, or other means of finding out their depositional attitude are available, it is not possible to determine the amount of post-formational tilting which has occurred. It is quite possible that some of the pole positions are in error for this reason. Thus one would expect the positions estimated from the lavas of Antrim and Mull to be closer to each other in view of their similar age. The Antrim lavas were corrected for dip (1° or 2°) while in the case of Mull dip corrections were not applied or even discussed, though dips of the order of 20° do occur.

(d) The secular variation has not been averaged out in some instances. It is likely that in a succession of lava flows, or a large intrusive body which has been adequately sampled, the effect of secular variation will have been averaged out. This may not be so in the case of the dykes which have been studied, as the time taken for their formation could be quite small compared with the period of secular variation.

It is quite clear that there is need for much more work on Tertiary rocks. A study of Skye was undertaken for the following reasons:

(i) To provide further data on the position of the pole during the Tertiary. Skye and Greenland are the two outstanding parts of the Thulean igneous province with large volumes of lava whose magnetization directions have not been studied.

(ii) To provide more information regarding reversals of the Earth's magnetic field. If, as some workers feel, reversals of the field occurred at regular intervals during the Tertiary, then normally and reversely magnetized rocks should be roughly equally abundant. As much information as possible is needed to decide whether or not this is the case.

(iii) To see if any information on the Tertiary geology of the island could be provided (see Section 2).

1365246xb, 1960, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1960.tb00064.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

For comparison with the gabbros of Skye, it was decided to investigate gabbros from Rhum and Ardnamurchan. The magnetic properties of gabbros have not been as extensively studied as those of basalts and red sandstones. It is felt that gabbros could give useful magnetic information (Green & Jaeger 1956) and that further information of their magnetic properties would be useful.

## 2. Geology and sampling

(a) *Skye.* The Tertiary geology of Skye is shown diagrammatically in Figure 1 on which the sampling sites are also indicated. A series of lavas with individual flows ranging in thickness from 6 to 50 ft occupy most of the North-western half of the island. They are believed to form five groups of lavas, whose relationships are probably as shown in Figure 1(b) (Geological Survey, private communication). Groups 1 and 2 may be of the same age. Three orientated samples were taken from the generally massive central part of each lava from the localities shown in Table 2.

**Table 2**

*The lavas of Skye* (See Figures 1(a) and 1(b))

| Group | Rock types | Sampling locality | Ref. to Fig. 1 | No. of flows present | No. of flows sampled | Remarks |
|---|---|---|---|---|---|---|
| 1 | Mainly non-porphyritic olivine basalts | The Storr | A | 24 | 23 | Flow 7 thin and weathered |
| | | The Quirang | B | 8 | 8 | |
| | | Near Portree | C | 2 | 2 | |
| 2 | Porphyritic and non-porphyritic olivine basalts | Oisgill Bay | D | 8 | 8 | |
| 3 | Irregular alternations of mugearites and basalts | Beinne na Greine | E | 10 | 9 | Flow 9 poorly exposed |
| 4 | Mugearites and trachytes | Ros a Mheallain | F | 9 | 7 | Badly weathered stream section. Flows 2 & 5 not sampled. |
| 5 | Mugearites and basalts | Osdale River | G | ? | 1 | Poorly exposed group. Confused by many dykes. |

The basal two flows of group 1 were sampled near Portree but the number of flows present between these and those at the Storr is not known. The lavas of this group were sampled at both the Quirang and the Storr to see if it is possible to effect a correlation by comparing the patterns of secular variation revealed by the two sections.

The extrusion of lavas was followed by the intrusion of a plutonic complex now exposed in the southern part of the island. The Cuillins and Blaven are composed of gabbro which has the form of a large sheet or laccolith of complex composition. There appear to have been two phases of intrusion of gabbroic

13652465b, 1960, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1960.tb00064.x by Curtin University Library, Wiley Online Library on [27/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

magma between which was an intrusion of peridotite magma (Douglas Weedon, private communication). The earlier gabbro is unbanded and was sampled in Coire na Banachdich (locality H, Figure 1(a)) along with a xenolith of basalt in the gabbro. The later gabbro is beautifully banded and was sampled at Druim Hain (I).

The last phase in the cycle of igneous activity was the intrusion of a large number of basic dykes which were sampled at the Storr, Oisgill Bay, Osdale and Coire na Banachdich.

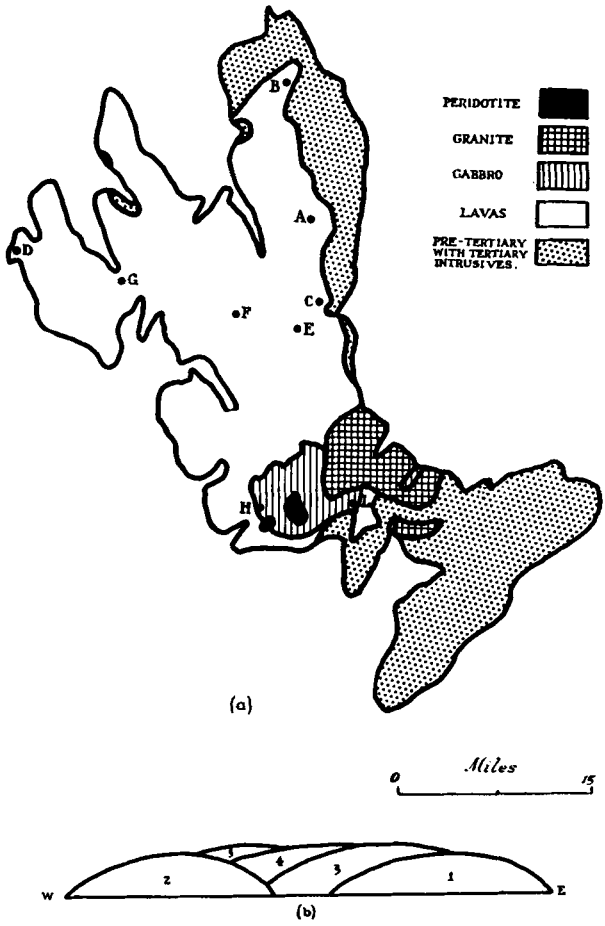



FIG. 1

(a) Geological sketch-map of Skye showing sampling localities A, B, C, ... etc., referred to in the text. (Reproduced from Sheet 1 of the 1/625 000 geological map of Great Britain by kind permission of the Controller, H.M. Stationery Office).
(b) Possible relationship of Skye lava groups.

(b) *Ardnamurchan.*—This area has been through a cycle of igneous activity similar to that of Skye but its plutonic history has been rather more complex. The plutonic rocks form a central intrusion ring complex, now in a fine state of preservation. The ring of banded hypersthene gabbro has been studied in detail by M. K. Wells (1953) and samples were taken from the following three localities described by him: (i) near Sanna Point; (ii) just south of the summit of Beinne na Seilg; and (iii) near Ardnamurchan Point.



13652465b, 1960, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1960.tb00064.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

(c) *Rhum.*—In the Hallival–Askival area of Rhum some 2 600 ft of rhythmically layered gabbro are exposed (Brown 1957). Each layer consists of a peridotitic rock which passes gradually upwards into an alluvialitic rock. Fifteen major rhythms varying in thickness (average 200 ft) have been recognized and the more resistant, well-exposed, alluvialitic parts of the top eight layers have been sampled.

## 3. Specimen preparation and measurement

In the laboratory the samples were set as *in situ* in plaster of Paris and up to six cylindrical cores of 1·9 cm diameter removed with a diamond impregnated bit fitted on to a standard drilling machine. The cores were then sliced into cylinders 1·9 cm long. These oriented cylinders were wedged into a cubical perspex box and the direction and intensity of the remanent magnetization measured on the Birmingham "spinning magnetometer" (Griffiths 1955). The error in direction of a single measurement made on the instrument is $\pm 1°$.

## 4. Stability criteria

In this work several specimens from each geological unit* have been subjected to one or more of the following tests to decide whether or not the magnetization of the unit is stable.

(a) *Statistical distribution of remanence directions.*—The directions of remanent magnetization of several specimens from each unit have been measured. The magnetization is considered to be stable if the directions of the specimens show a close grouping around a mean direction which is significantly different from that of the present Earth's field.

(b) *Change of direction of magnetization on storage.*—Most specimens were stored for six weeks after measurement with their directions of magnetization at right angles to the Earth's field in the laboratory and then remeasured. Specimens showing little change of direction (less than 3°) were regarded as stable, while those showing large changes were regarded as unstable (Thellier 1951, Blundell 1957).

(c) *NRM/TRM Ratio.*—The ratio of the intensity of the natural remanent magnetization (NRM) to that of the thermoremanent magnetization (TRM) produced by cooling from above the Curie Point in the Earth's field in the laboratory was estimated for at least one specimen from each unit. Specimens with the ratio less than but close to unity were regarded as stable; those for which the ratio is small (less than 0·2) were regarded as unstable; those with the ratio greater than unity are believed to have been affected by lightning (see below).

(d) *Thermal demagnetization.* The intensity of the NRM of rocks decreases with increase in temperature in a manner depending on the stability of the magnetization. Thus, unstably magnetized rocks show a rapid decrease of intensity as the temperature is increased to 50°C or 100°C; while the stably magnetized specimens show the main intensity drop at temperatures much nearer the Curie Point. A small number of thermal experiments was carried out using a non-magnetic furnace designed by Leng at Imperial College, London (Leng 1955). Curves of

* A geological unit is here taken to mean a body of igneous rock which has cooled from above its Curie Point to the air temperature in a time short compared with the period of secular variation (500–1000) years (Hospers 1953). Thus, a lava flow, a group of similar dykes at a site, or a small volume within a large intrusion is treated as a geological unit.

13652465b, 1960, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1960.tb00064.x by Curtin University Library, Wiley Online Library on [27/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

intensity against temperature were obtained during the heating and cooling of the specimen. The heating and cooling curves for both stable and unstable types of specimen are shown in Figure 2(a) and (b) (Nagata 1953).

(e) *A.C. demagnetization.*—The NRM of rocks is generally regarded as consisting of a "hard" (TRM) component of high coercivity, and a "soft" (isothermal remanent magnetization or IRM) component of low coercivity acquired through exposure for long periods of time in the Earth's magnetic field. An indication of stability can be obtained by noting the decrease in intensity of magnetization obtained by subjecting specimens to an alternating field which is smoothly decreased

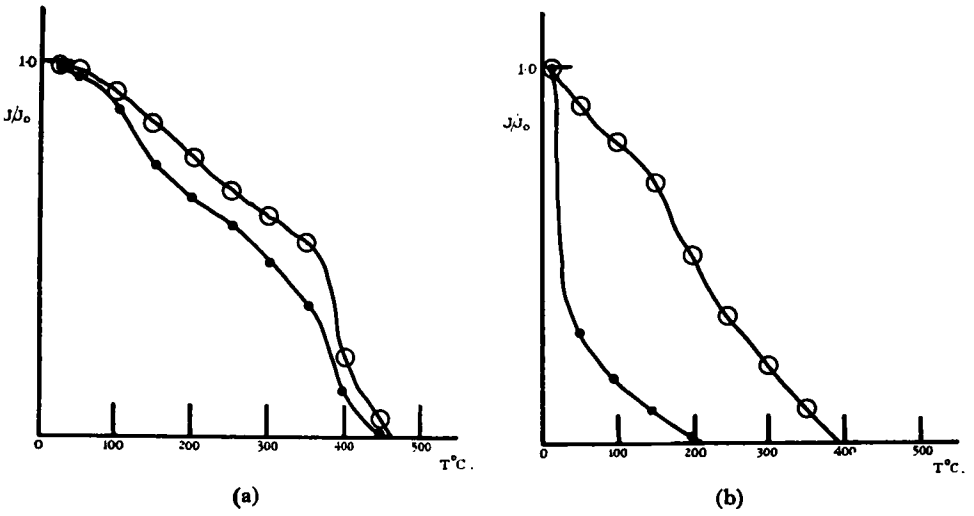



FIG. 2

Ordinate: normalized intensity of magnetization.
Abscissa: temperature (°C)

● heating; O cooling; J intensity of magnetization; $J_0$ intensity of magnetization at room temperature.

(a) Thermal demagnetization curves for a stable specimen.
(b) Thermal demagnetization curves for an unstable specimen.

to zero. The form of the curve of intensity of magnetization against the maximum applied alternating field can give a useful indication of stability (Uyeda 1958). Some workers (Creer 1958, As & others 1958) have successfully used alternating fields to remove unstable components from partially stable rocks. An attempt was made in this work to develop an apparatus to achieve such results.

To avoid the development of anhysteretic magnetization (Rimbert 1958) the experiments were carried out in a region in which the Earth's magnetic field was compensated by three Helmholtz pairs mutually at right angles. To allow for inadequate compensation and for any asymmetry in the wave form of the applied alternating field, the specimens were placed in a specially designed holder during demagnetization. The holder was rotated about two mutually perpendicular axes simultaneously while the field was reduced smoothly to zero using a liquid resistance. This arrangement works well for typical rock specimens to which a "hard" component and a "soft" component have been imparted in the laboratory. This is illustrated in Figure 3. The "hard" component was produced by cooling the specimen in the Earth's field from above the Curie Point to room temperature.

13652465b, 1960, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1960.tb00064.x by Curtin University Library, Wiley Online Library on [27/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The direction and intensity of this magnetization is represented by the points numbered 1. The specimen was then given a "soft" component by placing it in a direct field of 100 oersteds (points 2). The direction and intensity after each application of alternating field are shown by points 3 to 7. It is seen that the directions move along a great circle to that of the TRM when a peak alternating field of 100

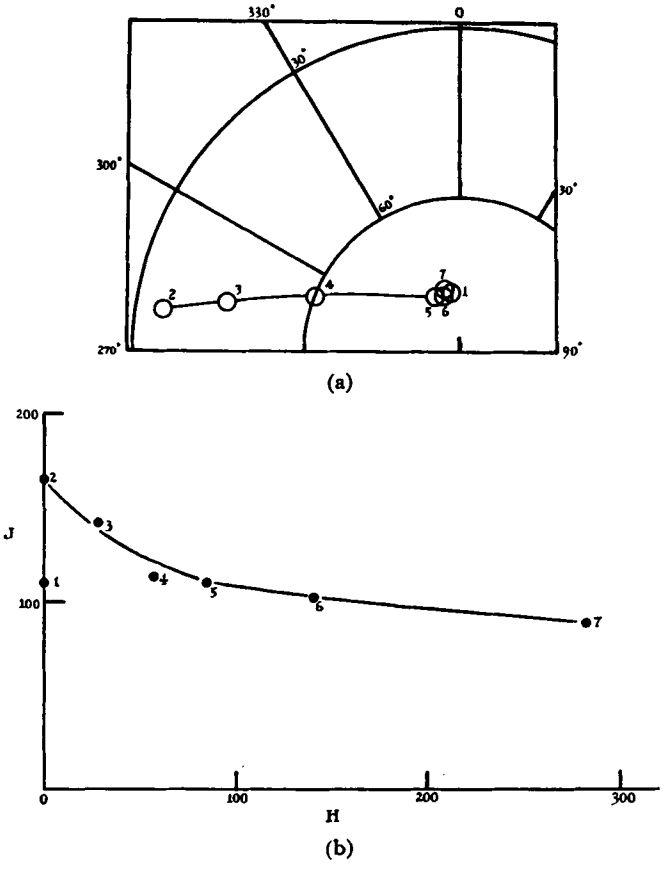



FIG. 3

A.c. demagnetization of an artificial two-component system. 1, thermoremanent magnetization produced in the laboratory; 2, thermoremanent magnetization plus an isothermal component produced by a direct field of 100 oersteds; 3, 4, 5, 6, 7 after application of an alternating field.

(a) Stereographic projection showing directions.
(b) Ordinate: intensity of magnetization (arbitrary units).
Abscissa: maximum applied alternating field (peak values in oersteds).
Variation of intensity with applied alternating field.

oersteds is reached and that the intensity of the TRM is also reached at this field. At higher fields, no change in direction is observed while the intensity of magnetization decreased steadily.

There is little doubt that in the above experiment the "soft" component of magnetization has been successfully removed. However, no specimen with an NRM behaved in this fashion. Figure 4 shows the directions of magnetization of 4 specimens from a single typical sample, which was thought to be partially stable, after successive applications of alternating fields as above. There is no

1365246x, 1960, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1960.tb00064.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

systematic change of direction. The intensities of magnetization also show no systematic behaviour and increases of up to 200 per cent as well as decreases have been observed after the application of fields up to 400 oersteds (r.m.s.).

It seems that the NRM of these specimens is much more complex than was envisaged and that its true nature must be the subject of further work.

Using the above stability criteria, it is possible to divide the specimens into three groups as follows:

(a) A stable group in which the magnetization directions from a unit are closely grouped around a mean direction nearly opposite that of the present Earth's field (Figure 5) and do not change appreciably on storage. The NRM/TRM ratio is on the average 0·7 to 0·8, and the thermal demagnetization curves are of the stable type. They show little or no change of direction and some loss of intensity when subjected to an alternating field of 400 oersteds (r.m.s.). There is little doubt that the NRM of these specimens has been acquired by cooling through their Curie Points in the Earth's magnetic field. Their magnetization directions thus reflect that of the Earth's field acting when the magnetization was acquired.

(b) An unstable group in which the magnetization directions of the specimens from a unit are widely dispersed (Figure 6) and change by up to 90° on storage. The NRM/TRM ratios are generally small (down to 0·01), and the thermal demagnetization curves are of the unstable type. The magnetization is greatly affected by quite low alternating fields, sometimes a decrease of the order of 50 per cent in intensity is produced by a field of 10 oersteds. The magnetization of such specimens are clearly unreliable and they are therefore of no use for palaeomagnetic purposes.

(c) In the third group, the specimens from within a sample (roughly a 4 in. to 6 in. cube) usually show a close grouping of directions which are not altered on storage. The mean direction of samples from within a unit are, however, quite different (Figure 7). The NRM/TRM ratios are usually greater than unity, values ranging from 1·2 to 35·0 having been observed, and the thermal demagnetization curves are of the stable type. Alternating fields of up to 400 oersteds have little effect on the magnetization. All samples showing these properties have been collected from the tops of hills. They have evidently been magnetized in fields other than that of the Earth, and it seems likely that the strong alternating and direct fields associated with lightning discharges have led to the development of anhysteretic remanent magnetization in these samples. Such samples are obviously of no use for palaeomagnetic purposes and have therefore been neglected.

## 5. Results

The results of the measurements made are summarized in Table 3 in which the total number of specimens measured and the numbers rejected because of instability and lightning effects are given. The mean intensities of magnetization in e.m.u./cc for the stable, unstable and lightning-affected specimens are given for each locality. The mean declinations, inclinations, angles of confidence ($P = 33\frac{1}{3}$ per cent), and dispersion parameters were estimated weighting each unit equally.

The declination and inclinations of individual flows from the Storr and the Quirang are shown in Figure 8, the errors being the values of the semi-angle of the cone of $66\frac{2}{3}$ per cent confidence which gives the spherical equivalent of the standard deviation.

13652465b, 1960, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1960.tb00064.x by Curtin University Library, Wiley Online Library on [27/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

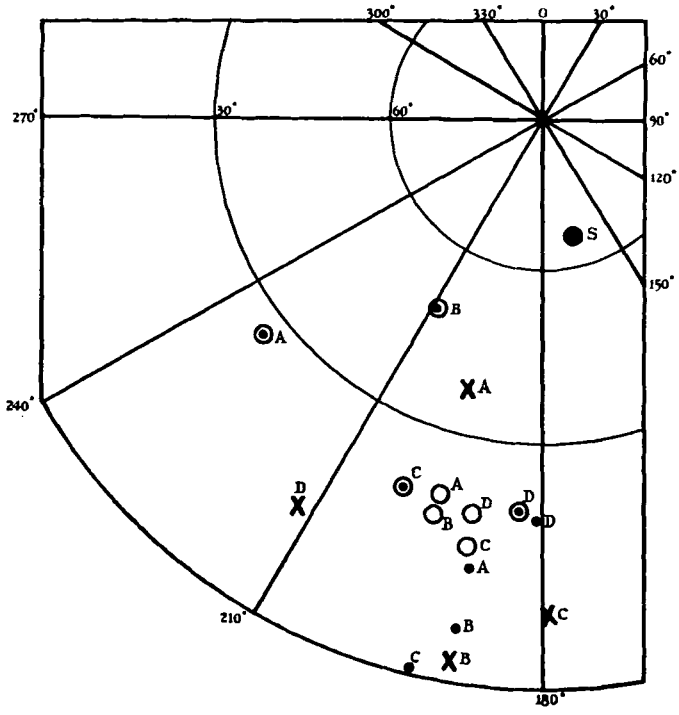



FIG. 4

Stereographic projection showing magnetization directions of specimens A, B, C, and D from the same sample after a.c. demagnetization. ⊙ directions of N.R.M.; ○ after 50 oersteds (r.m.s.); ● after 100 oersteds; × after 200 oersteds ● S present south pole (upper hemisphere, all other points are in the lower hemisphere).

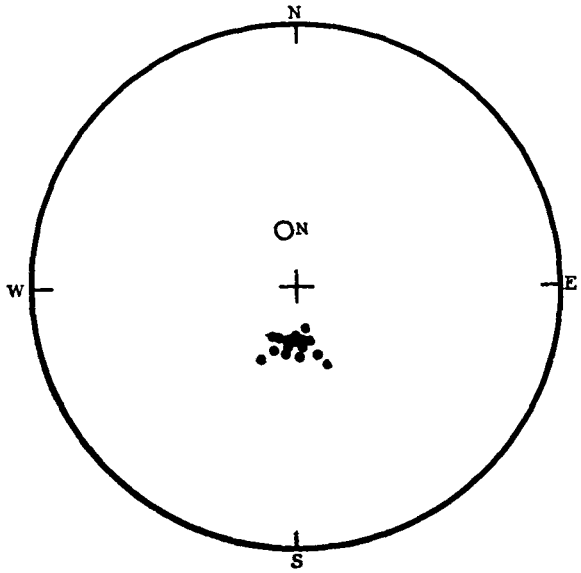



FIG. 5

Stereographic projection showing magnetization directions of specimens from a stably magnetized geological unit. ○N present North pole; ● points in upper hemisphere.

13652465b, 1960, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1960.tb00064.x by Curtin University Library, Wiley Online Library on [27/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

## Table 3

*Summary of results*

| Rocks | Total Number of specimens | Number of unstable specimens | Number of lightning-affected specimens | Mean intensity of stable specimens e.m.u./cc ($\times 10^{-3}$) | Mean intensity of unstable specimens e.m.u./cc ($\times 10^{-3}$) | Mean intensity of lightning-affected specimens e.m.u./cc ($\times 10^{-3}$) | Number of reliable geological units | Mean declination (degrees from true North) | Mean inclination (degrees from horizontal) | Semi-angle of cone of 66⅔ per cent confidence | Dispersion parameter |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Storr lavas (no dip correction) | 230 | 11 | 16 | 4·01 | 1·40 | 46·25 | 21 | 191·0 | 61·6 | 2·9 | 44·5 |
| Storr lavas (after dip correction) | 230 | 11 | 16 | 4·01 | 1·40 | 46·25 | 21 | 162·0 | 62·0 | 2·9 | 44·5 |
| Quirang lavas | 75 | 4 | — | 4·02 | 0·33 | — | 8 | 183·0 | 57·3 | 4·5 | 57·3 |
| Oisgill Bay lavas | 103 | 39 | 6 | 5·75 | 0·80 | 1·49 | 7 | 190·3 | 62·3 | 7·1 | 23·2 |
| Beinne na Greine lavas | 106 | 42 | 24 | 3·79 | 1·52 | 50·85 | 6 | 168·2 | 54·0 | 8·8 | 17·9 |
| Bracadale lavas | 83 | 55 | 9 | 1·74 | 2·02 | 7·50 | 3 | 187·8 | 52·4 | 6·2 | 84·8 |
| Osdale lavas | 13 | — | — | 3·54 | — | — | 1 | 166·6 | 61·5 | — | — |
| Portree lavas | 14 | — | 5 | 2·21 | — | 19·97 | 1 | 197·6 | 67·6 | — | — |
| Coire na Banachdich gabbro | 24 | — | — | 4·55 | — | — | 1 | 199·1 | 65·8 | — | — |
| Coire na Banachdich basalt xenolith | 5 | — | — | 2·58 | — | — | 1 | 181·9 | 73·6 | — | — |
| Storr dykes | 22 | 14 | — | 2·13 | 0·48 | — | 1 | 183·0 | 41·0 | — | — |
| Oisgill dykes | 13 | — | — | 2·54 | — | — | 1 | 187·7 | 44·0 | — | — |
| Coire na Banachdich dykes | 4 | — | — | 3·24 | — | — | 1 | 212·1 | 63·4 | — | — |
| Druim Hain gabbro | 11 | — | 11 | — | — | 2·97 | — | — | — | — | — |
| Osdale dykes | 21 | — | — | 1·02 | — | — | 1 | 181·9 | 56·0 | — | — |
| All Skye | 724 | 165 | 71 | 3·81 | 1·39 | 30·55 | 53 | 186·1 | 59·7 | 2·0 | 59·7 |
| Ardnamurchan hypersthene gabbro | 65 | — | 24 | 4·08 | — | 29·61 | 3 | 184·1 | 54·5 | 7·2 | 61·9 |
| Rhum layered gabbro | 38 | — | 11 | 1·22 | — | 136·33 | 4 | 181·0 | 55·1 | 4·6 | 97·1 |

13652464b, 1960, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1960.tb00064.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

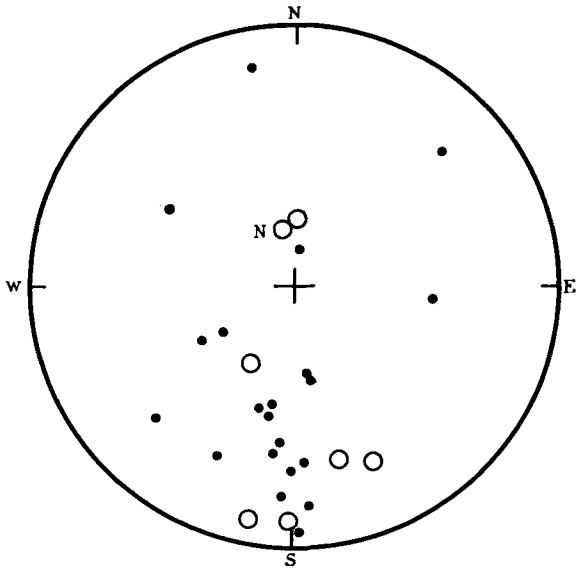



FIG. 6.—Stereographic projection showing magnetization directions of specimens from an unstably magnetized geological unit. ○N present North Pole; ● points in upper hemisphere; ○ points in lower hemisphere.

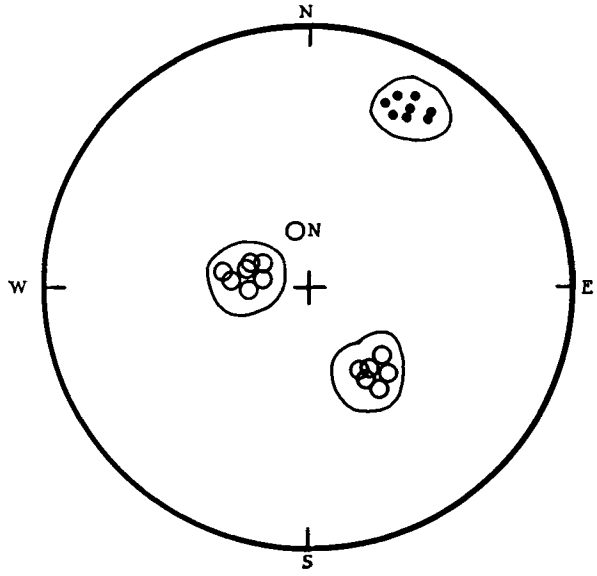



FIG. 7.—Stereographic projection showing magnetization directions of specimens from three samples of a unit affected by lightning. Specimens from the same sample are enclosed by closed curves. ○N present North Pole; ○ points in lower hemisphere; ● points in upper hemisphere.

13652465b, 1960, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1960.tb00064.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The mean direction of magnetization of each of the Skye units is shown in Figure 9.

The ancient pole positions estimated from Skye, Ardnamurchan and Rhum in present-day geographical coordinates are given in Table 4.

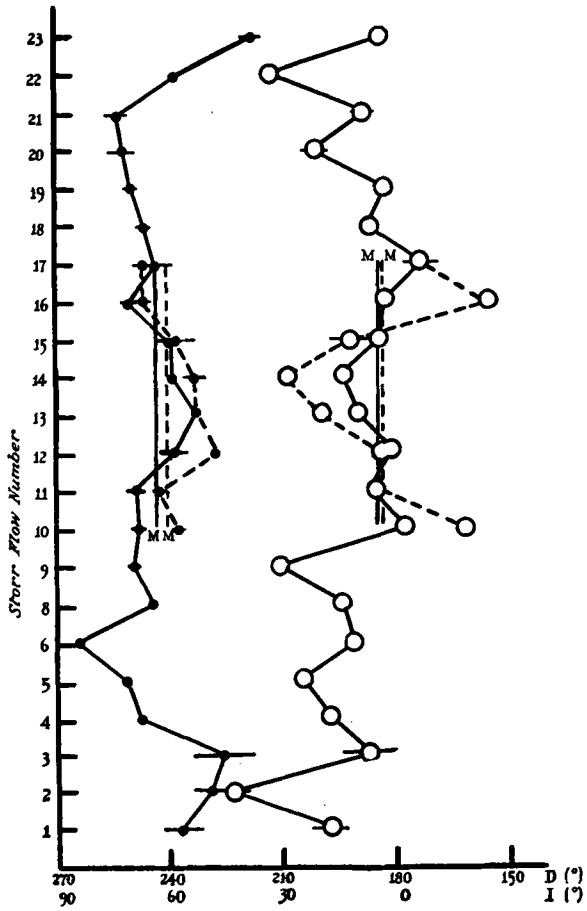



FIG. 8. Secular variation from the lavas of the Storr and Quirang. ● inclination; ○ declination; ——Storr; - - - - Quirang; M mean.

## Table 4

*Ancient pole positions*

| Location | Ancient Pole Position Lat. | Long. | *dp* | *dm* |
|---|---|---|---|---|
| Skye | 70°·8 N | 159°·6 E | 3°·8 | 5°·1 |
| Ardnamurchan | 68°·1 N | 165°·0 E | 15°·4 | 21°·9 |
| Rhum | 68°·5 N | 150°·9 E | 14°·0 | 13°·4 |

The half angles of the oval cones of confidence are here estimated at the 95 per cent level for easy comparison with the results of earlier workers given in Table 1.

13652465b, 1960, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1960.tb00064.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

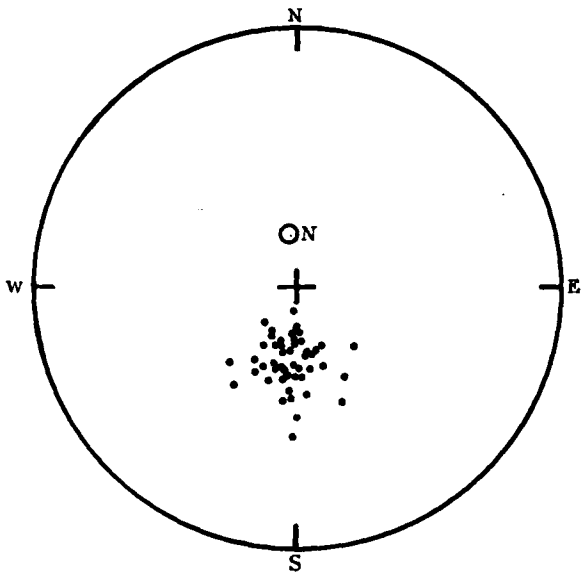



FIG. 9. Stereographic projection showing magnetization directions of each of the Skye geological units. ○N present North Pole; ● points in upper hemisphere.

## 6. Discussion of results

It is seen from Table 3 that about 10 per cent of the Skye rocks (all from hill-tops) have been affected by lightning and about 20 per cent possess unstable magnetization. In some instances, e.g. most of the specimens from the stream section near Bracadale, the rocks possessing unstable magnetization are more weathered than the average. It is also evident that, in general, the unstable rocks have lower intensities of magnetization than the stable ones, while those affected by lightning have much higher intensities. None of the gabbros was found to be unstable but about 40 per cent were affected by lightning.

The lavas at the Storr have a westerly dip of 14° (determined with the aid of orientated photographs). The base of the lavas all round the island is not far from sea level and Dr F. W. Anderson of the Geological Survey has estimated that the lavas have not been tilted by more than ¼° since they were formed. This is supported by the fact that the lavas of Oisgill Bay, which are horizontal, and are probably of the same age, have the same mean magnetization direction as the Storr lavas provided they are not corrected, but the difference is large if a correction is made. It is, therefore, concluded that no dip corrections are necessary for the Skye rocks.

The declinations and inclinations of the individual flows of the Storr and the Quirang are shown in Figure 8. The differences in magnetization direction between the flows are attributed to secular variation. The maximum departure is seen to be within 30° of the mean in declination and 25° in inclination, values which are compatible with the views of other workers. An attempt was made to correlate the lavas of the two sections by comparing their secular variation patterns by eye; the best fit was obtained by equating the basal Quirang flow with the tenth flow of the Storr. In addition, for each possible correlation (assuming that all the

13652465b, 1960, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1960.tb00064.x by Curtin University Library, Wiley Online Library on [27/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Quirang flows are represented at the Storr) two correlation coefficients were estimated—one for the eight pairs of declination values, and one for the eight pairs of inclination values. The fifteen pairs of coefficients, obtained by equating the basal Quirang flow with each of the lower fifteen Storr flows in turn, are plotted in Figure 10 as ordinates against the Storr flow number with which the basal Quirang

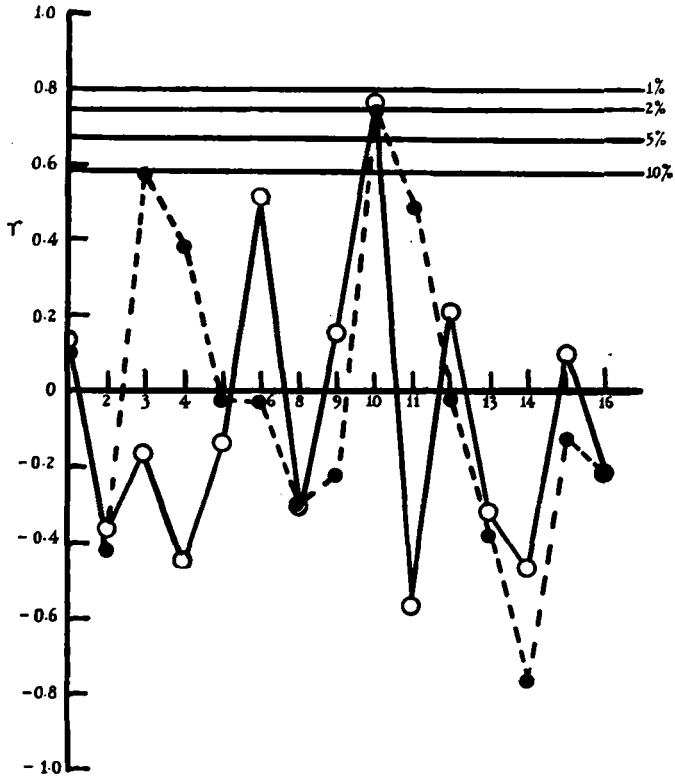



FIG. 10

Ordinate: coefficient of correlation
Abscissa: Storr flow number with which the basal Quirang flow is correlated.

Correlation of lavas of flows of the Storr and the Quirang. ——inclinations; - - - - declinations. The 1, 2, 5 and 10 per cent levels of significance are shown.

flow is correlated. For both parameters the maximum value of the coefficient is obtained when the basal Quirang is correlated with the tenth flow of the Storr, and it is seen that in both there is only a 2 per cent probability that this is due to chance. It is also noticeable that this is the only postulated correlation for which the correlation coefficient is of a value usually regarded as significant (Paradine & Rivett 1953). The mean declination and inclination of the two equivalent sets of flow are seen to be almost identical. The above is quite strong evidence for concluding that if there is complete overlap of the two sets of flows, the basal Quirang flow must be equated with the tenth Storr flow.

It is not possible to effect a similar correlation between the lavas of the Storr and Oisgill Bay as they belong to different groups which probably do not contain flows which were formed exactly contemporaneously.

13652465b, 1960, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1960.tb00064.x by Curtin University Library, Wiley Online Library on [27/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The differences between the mean magnetization directions for the various sites given in Table 3 are attributed to incomplete averaging of the secular variation. The similarity of the magnetization directions shown by the dykes of the Storr, Oisgill Bay, and Osdale probably means that they were formed at the same time, which is different from that at which the Coire na Banachdich dykes were formed.

A striking fact about the directions of magnetization is that all the stably magnetized rocks show reverse magnetization. As none of the rocks shows self-reversing properties in the laboratory, the reverse magnetization is believed to be due to the rocks being formed when the Earth's magnetic field was reversed. It is difficult to estimate the time covered by this. Hawkes (1916) estimated that the average time between flows in Iceland is about 1 000 years, but Hospers (1954) argues that this is probably an underestimate, and suggests a time nearer 10 000 years. Since, for example, in Skye there are over 50 flows, the Earth's field must have been reversed for a time of the order of half a million years.

It is highly probable that by taking the mean of the fifty-three Skye units examined, the effect of secular variation will be averaged out. The ancient pole position estimated from this mean therefore corresponds to that of the geographic pole position at the time of formation of the rocks, i.e. probably Lower Miocene. This position (Table 4) is significantly different from that of the present geographic pole. It therefore seems that there has been some relative movement between North-west Europe and the pole since this time. There is still a great need, however, for further results from other Tertiary areas (e.g. Greenland, Spitzbergen, Jan Mayen, The Faroes). This, along with further data from America and accurate dating (e.g. by the Argon method) would provide important information about relative movement between the two continents.

The gabbros from Rhum and Ardnamurchan (excluding the lightning-affected samples) were found to be reversely magnetized, but not enough sites were sampled to give an accurate estimate of the pole position in either case. It is interesting to note, however, that they were found to be eminently suitable for palaeomagnetic purposes (Blundell & Read 1958, Gough & Niekerk 1959). There is scope for much more work on gabbros, for by suitable sampling and thermal studies in the laboratory, they can provide a more continuous pattern of secular variation than can a succession of lava flows (Jaeger 1957). The pole positions estimated (Table 4) from Rhum and Ardnamurchan are quite close to that estimated from Skye, and the fact that they are also reversely magnetized and that no normally magnetized rocks were found, could mean that the rocks studied from all three areas were formed within the same period of reversal of the field, i.e. within the same half million years.

## 7. Acknowledgments

The writer is grateful to Professor J. McG. Bruckshaw and Dr D. H. Griffiths for suggesting this work, which has been carried out with the constant help of Dr R. F. King and Dr Griffiths.

He is also grateful to many others whose help made this work possible, especially Dr F. W. Anderson and Dr J. B. Simpson of the Geological Survey of Great Britain and Dr Douglas Weedon of the University of Glasgow for access to unpublished geological works; Dr J. A. Clegg and Dr C. W. F. Everitt of the Physics Department, Imperial College of Science, London for generous use of and kind assistance with the non-magnetic furnace; Dr A. I. Rees whose help in the field

13652465b, 1960, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1960.tb00064.x by Curtin University Library, Wiley Online Library on [27/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

has been invaluable; Professor F. W. Shotton and other members of the Geology Department of the University of Birmingham, for generous use of departmental facilities; and Mr S. Bright for invaluable technical assistance.

Finally, he wishes to thank the Trinidad Petroleum Development Company for an award during the first half of the investigation and to the University of Birmingham for a grant during the second half.

*Department of Geology,*
*University of Birmingham,*
1959 *August.*

## References

As, J. A. & Zijderveld, J. D. A., 1958. *Geophys. J.*, **1**, 308.
Baily, W. H., 1869. *Quart. J. Geol. Soc. Lond.*, **25**, 357.
Blundell, D. J., 1957. *Geol. Mag.*, **94**, 187.
Blundell, D. J. & Read, H. H., 1958. *Proc. Geol. Ass.*, **69**, 191.
Brown, G. M., 1957. *Phil. Trans. Roy. Soc. B.*, **240**, 1.
Bruckshaw, J. M. & Robertson, E. I., 1949. *Mon. Not. R. Astr. Soc. Geophys. Suppl.*, **5**, 308.
Creer, K. M., 1958. *Ann. Geophys.*, **14**, 393.
Dutt, S. K., 1955. Thesis: Univ. of London.
Fisher, R. A., 1953. *Proc. Roy. Soc. A.*, **217**, 295.
Forbes, E., 1851. *Quart. J. Geol. Soc. Lond.*, **7**, 103.
Gardner, J. Starkie, 1885a. *Quart. J. Geol. Soc. Lond.*, **41**, 82.
Gardner, J. Starkie, 1885b. *Quart. J. Geol. Soc. Lond.*, **41**, 93.
Gignoux, M., 1955. *Stratigraphic Geology*, California.
Gough, D. I. & van Niekerk, C. B., 1959. *Phil. Mag.*, **4**, 126.
Green, R. & Jaeger, J. C., 1956. *Geofisica pura e app.*, **35**, 49.
Griffiths, D. H., 1955. *Mon. Not. R. Astr. Soc. Geophys. Suppl.* **7**, 103.
Hawkes, L., 1916. *Geol. Mag.*, **53**, 385.
Hawkes, L., 1938. *Geol. Mag.*, **75**, 289.
Harker, A., 1904. Tertiary Igneous Rocks of Skye. *Mem. Geol. Surv.*
Heer, O., 1868. *Flora Fossils Artica*, Zurich.
Hospers, J., 1953. *Kon. Ned. Akad. Wet., B.*, **56**, 467.
Hospers, J. & Charlesworth, H. A. K., 1954. *Mon. Not. R. Astr. Soc. Geophys. Suppl.*, **7**, 32.
Irving, E., 1956. *Geofisica pura e app.*, **33**, 23.
Irving, E., 1957. *Phil. Mag. Suppl.*, **6**, 194.
Irving, E., 1959. *Geophys. J.*, **2**, 51.
Jaeger, J. C., 1957. *Amer. J. Sci.*, **255**, 306.
Leng, J. H., 1955. Thesis: University of London.
Nagata, T., 1953. *Rock Magnetism*, Tokyo.
Paradine, C. G. & Rivett, B. H. P., 1953. *Statistics for Technologists*, London.
Reid, C., 1913. (In) Geology of the country round Newton Abbot, *Mem. Geol. Surv.*
Rimbert, F., 1958. Thesis: Paris.
Roche, A., 1950a. *C.R. Acad. Sci. Paris.*, **230**, 113.
Roche, A., 1950b. *C.R. Acad. Sci. Paris*, **230**, 1603.

13652465x, 1960, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1960.tb00064.x by Curtin University Library, Wiley Online Library on [27/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Roche, A., 1951. *C.R. Acad. Sci. Paris*, **233**, 1132.
Seward, A. C. & Holtum, R. E., 1924. (In) Tertiary and Post-Tertiary Geology of Mull, Loch Aline, and Oban. *Mem. Geol. Surv. Scotland.*
Simpson, J. B., 1951. British Assoc. Programme, Section C, 63.
Thellier, E., 1951. *J. Phys. Rad.*, **12**, 205.
Uyeda, S., 1958. *Jap. J. Geophys.*, **II**, 1.
Vincenz, S. A., 1954. *Mon. Not. R. Astr. Soc. Geophys. Suppl.*, **6**, 590.
Wells, M. K., 1953. *Quart. J. Geol. Soc. Lond.*, **109**, 367.

13652465b, 1960, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1960.tb00064.x by Curtin University Library, Wiley Online Library on [27/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License